Automated eDNA extractions of filtered water samples: Kingfisher Apex with MagMax chemistry

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Background

This has been adapted from the MagMax Multi-sample Ultra 2.0 manufacturer's protocol for 96-well Kingfisher format to allow for the increased lysate volume that's necessary for Sterivex filters.

Important Considerations

- This protocol is written for use of Sterivex filters, which require a large lysate volume compared to smaller syringe filters or what the standard kit protocol calls for. You can adapt this protocol to larger or smaller lysates by scaling the amount of lysis buffer, proteinase K, and lysis/binding solution you use. If scaling down, you can likely decrease the number of input blocks required as well. Contact Katie Murphy with questions.
- Make sure you've incorporated negative controls into all stages of your experiment (e.g. filtration blank, extraction blank).
- Make a plate map ahead of time so that you know what sample will go into what well on the blocks/plates.
- Up to 96 samples can be extracted at a time on the Kingfisher.
- The on-instrument script only takes about an hour to run, but keep in mind there will be significant setup time beforehand.
- Decide what elution volume you wish to use (100, 150, or 200 µL); make sure to use the corresponding Kingfisher script.

Materials & Reagents

- MagMax DNA Multi-Sample Ultra 2.0 reagents (bulk catalog #s listed below; can also use the kit):
    - Binding Beads (Thermo Fisher catalog #A36579)
    - Wash I solution (Thermo Fisher catalog #A36580)
    - Lysis/binding solution (Thermo Fisher catalog #A36581)
    - Elution solution (Thermo Fisher catalog #A36582)
        - Alternatively, use 10 mM Tris or TLE
- 20 mg/ml Proteinase K
- Storage/lysis buffer: Longmires, DNA/RNA Shield
- 100% ethanol
- Nuclease-free water
- 3 mL Luer-lock syringes
- 9 Deep 96-well blocks for Kingfisher (barcoded preferred)
- 96-well tip comb for Kingfisher
- Sterivex filters containing your eSamples.
- Outlet caps – e.g. Cole-Parmer Female Luer Fitting EW-30800-12
    - Only needed if your sterivex aren't already capped with these
- Inlet caps – e.g. Cole-Parmer Male Luer Fitting EW-30800-30
    - Only needed if your sterivex aren't already capped with these
- Parafilm
- 10% bleach
- 70% EtOH
- Filter tips

- Sterile PCR plates
- *Optional:* Internal positive controls (IPCs)

Protocol

Day 1:

1. Clean all surfaces, tip boxes, racks, pipettes and so on with 10% bleach followed by 70% EtOH.
2. Make sure all sterivex and tubes are thoroughly labeled.
3. Add an outlet cap (narrow plug) to each sterivex if it doesn't already have one. Push firmly but carefully into place.
4. Place each sterivex upright in a rack (15 mL tube racks work well). Include a 5 mL tube to serve as an extraction blank.
5. If your sterivex are currently empty (no storage buffer), add 1.76 mL of storage/lysis buffer via the inlet port using either a P1000 with filter tips or a 3 mL syringe. Use a new tip or syringe for each sample.
    a. If your sterivex already contain a lower volume of storage/lysis buffer, bring the volume up to approximate 1.76 mL.
6. Add 1.76 mL of storage/lysis buffer to the 5 mL tube extraction blank.
7. Add 80 µL of MagMax Enhancer solution to each sterivex or blank via the inlet port.
8. Add 160 µL Proteinase K to each sterivex or blank via the inlet port.
9. *Optional:* if spiking in an internal positive control, you may wish to do so now.
10. Securely cap each sterivex using an inlet cap.
11. Invert each sterivex or tube 10X to mix.
12. Thoroughly seal around each inlet and outlet cap with parafilm to ensure they do not leak during the overnight incubation.
13. Place each sterivex or tube into a Rototherm. Incubate overnight at 56C with rotation speed set to 20 rpm.
    a. The sterivex and 5 mL tubes both fit perfectly into the blue Rototherm tube holders designed for 15 mL tubes.
14. When you are done for the day, clean all surfaces, tip boxes, racks, pipettes and so on with 10% bleach followed by 70% EtOH.

Day 2:

1. Clean all surfaces, tip boxes, racks, pipettes and so on with 10% bleach followed by 70% EtOH.
2. If you haven't already done so, make a plate map indicating what sample will go into what well on the blocks/plates.
3. Prepare fresh 80% EtOH using nuclease-free water. You will need 1.5 mL per sample.
4. Label 5 deep-well blocks as indicated below, then add the following contents to each well of the block using a 96-channel pipette or repeat pipettor. Only add reagents to the wells you need to use; leave unused wells empty to conserve reagents. *Note: these blocks can be prepared up to a day ahead of time; seal well and store at room temp if prepared ahead of time.*
    a. Wash I block: add 1 mL MagMag kit wash I solution to each well
    b. Wash II block 1: add 1 mL 80% EtOH to each well.
    c. Wash II block 2: add 0.5 mL 80% EtOH to each well
    d. Elution block: add 200 µL MagMax elution solution to each well.
        i. Can adjust to 100 or 150 µL if preferred.
    e. Tip block: set new tip comb into block.
5. Label four new 96-well blocks as Sample #1, #2, #3, and #4.
    a. *If you used less than 1.76 mL lysis buffer during Day 1 step 6, you may need to adjust this protocol and your Kingfisher script to allow for fewer sample blocks. Do not just follow the default. Contact Katie Murphy with questions.*

6. Add 400 µL **binding solution** (***no beads!)*** to Sample Blocks #2, #3, and #4; only add to the wells that will contain samples.
7. Transfer all **lysate** from each sterivex using 3 mL syringes into its corresponding well on Sample Block #1 (refer to your plate map).
8. Aliquot 500 µL of **lysate** from Sample Block #1 into Sample Blocks #2-4 using 96-channel pipette (or your preferred pipette).
    a. The wells in Sample blocks #2-4 will now each contain 900 µL total (400 µL binding solution and 500 µL of lysate).
9. Prepare the **bead binding mix** in a 15 mL or 50 mL conical; prepare enough for all of your reactions plus ~10% excess. Each reaction requires 400 µL of binding solution and 40 µL of beads. Add binding solution and beads to conical tube. Carefully invert to mix until solution appears homogenous.
10. Carefully add 440 µL **bead binding mix** to all wells of Sample Block #1.
    a. Final contents of Sample Block #1: approx. 500 uL lysate + 440 uL bead binding mix.
11. Power on the Kingfisher Apex (if not already on).
    a. There is a physical flip switch on the back right, followed by a button on the front right.
12. On the Kingfisher touchscreen, initiate the script that matches your chosen elution volume.
    a. The scripts are named:
        i. LAB_eDNA_ultra2_96_**200uL**_APEX
        ii. LAB_eDNA_ultra2_96_**150uL**_APEX
        iii. LAB_eDNA_ultra2_96_**100uL**_APEX
13. Carefully raise the lid on the Kingfisher Apex.
14. Following the on-screen prompts, insert your prepared blocks one at a time onto the platform's loading location.
    a. If your block have barcodes, the barcode should be facing backwards (towards the scanner).
15. Once all blocks have been loaded, lower the Kingfisher lid.
16. If required, click to proceed on the touchscreen.
17. Once initiated, the script run time is approximately 1 hr 10 minutes.
18. Once the script is completed, carefully remove all blocks from the deck following the on-screen prompts.
19. Seal your elution block with a foil seal; this contains your extracted eDNA.
20. Spin the elution block in the plate centrifuge at 400g for 1 min to collect contents to the bottom of the wells.
21. Label and date a new PCR plate (or preferred plate/tubes) as your eDNA stock plate.
22. Transfer the eluted DNA from the elution block into your new eDNA stock plate.
23. *Optional*: You may wish to make eDNA aliquot plates at this time too, to minimize the number of times you need to freeze/thaw plates and to reduce contamination.
24. Heat seal your eDNA stock plate as well as any eDNA aliquot plates. The heat sealer should be set to 175C for 3 seconds.
25. After sealing, your eDNA should be stored at -20C (or at 4C for immediate use).
26. Any spills/splashes on the Kingfisher deck should first be wiped up with water, then follow with 10% bleach and 70% EtOH. Dry deck thoroughly.
27. Close the Kingfisher lid and turn on the deck UV for 30 minutes.
28. Clean up the remaining reagent blocks from extraction. All liquid waste should be collected into a labeled waste bottle. All solid waste (including the blocks) should be double bagged. Both liquid and solid waste are stored below the sink in the eDNA lab. *Once a waste bag or bottle is full, it must be transferred to the regular satellite waste collection point in the main pre-PCR lab at MSC.*
29. Clean all other work surfaces, tip boxes, racks, pipettes and so on with 10% bleach followed by 70% EtOH.

References

MagMAX DNA Multi-Sample Ultra 2.0 Kit User Guide (tissue). MAN0018807 – rev B. July 2024. Thermo Fisher Scientific.